HU108 Test Pipeline — Logic & Verification Guide

HU108 测试流程 —— 逻辑与验证说明

1. Two Different Questions / 两个不同的问题

There are two completely different questions that are easy to confuse:
这里有两个很容易混淆的问题，需要先分清楚：

Question	Answered by
Does my code work?	Simulated test (--simulate)
我的代码能不能正常运行?	模拟数据测试
Is my sensor / vibration data good?	Real hardware test
我的传感器 / 振动数据好不好?	真实硬件测试

The --simulate mode only answers the first question. The data it produces is synthetic (made up by mock_serial.py), so asking "is this data good or bad" does not make sense yet — there is no real physical phenomenon behind it.
--simulate 模式只回答第一个问题。它产生的数据是编造出来的（在 mock_serial.py 里手写的），所以问“这个数据好不好”目前是没有意义的——它背后没有真实的物理现象。

2. What mock_serial.py Actually Generates / mock_serial.py 到底生成了什么

```
x = 0.05 * sin(2π * 3.0 Hz * t) + noise
y = 0.03 * sin(2π * 5.0 Hz * t) + noise
z = 0.02 * sin(2π * 1.5 Hz * t) + noise
# plus a brief "spike" roughly every 5 seconds
# 另外每隔约 5 秒加一个“冲击”尖峰
```

This is a known, designed signal — I chose these frequencies and amplitudes myself. The purpose is to have a signal with a known ground truth, so we can check whether the pipeline correctly captures and reproduces it.
这是一个已知的、人为设计的信号——频率和振幅是我自己定的。目的是让这个信号有一个“已知的标准答案”，这样才能检查整条流程有没有正确地把它采集、解析、画出来。

3. What This Test Actually Verifies (Software Pipeline) / 这次测试真正验证的内容（软件流程）

Check	How to verify	What it tells you
Sampling is steady, no major stalls 采样是否稳定、没有明显卡顿	Look at the timestamp column in the CSV — intervals should be close to the designed rate (~0.01s /	Whether the read loop has bugs 读取循环有没有问题（阻塞、丢

Check	How to verify	What it tells you
	100Hz) 看 CSV 的 timestamp 列，间隔应接近设计值(约 0.01 秒/100Hz)	帧）
Plot matches the designed signal 画出的波形是否符合设计	X should oscillate faster than Z (3Hz vs 1.5Hz); Y fastest (5Hz) X 轴应比 Z 轴跳动快(3Hz vs 1.5Hz)；Y 轴最快(5Hz)	Whether parse_line() maps columns correctly 解析函数有没有把 x,y,z 顺序搞混
Spike appears where expected 尖峰出现在预期位置	A visible jump near t≈0s (and every ~5s for longer runs) 在 t≈0 秒附近（以及每隔约 5 秒）应有明显跳变	Whether transient events are captured, not smoothed away 瞬态事件有没有被正确捕捉
Row count matches expected duration 数据行数是否符合预期时长	~duration × 100 rows (e.g. 3s → ~300 rows) 约 时长×100 行（例如 3 秒→约 300 行）	Whether samples are being silently dropped 有没有静默丢数据

Example check we already ran / 已经跑过的示例检查

```
样本数 (samples): 298
时间跨度 (duration): 3.009 秒 (s)
平均采样间隔 (avg interval): 0.0101 秒 (s)  — designed value / 设计值: 0.01s
```

This matched the design closely, confirming the pipeline logic (read → parse → CSV → plot) works correctly.
这个结果和设计值几乎一致，说明整条流程（读取→解析→存 CSV→画图）的逻辑没有问题。

4. What "Good Data" Means Once Real Hardware Is Connected / 接上真实硬件之后，“数据好不好”应该怎么判断

Once the HU108 is physically connected, the checks change from "does the code work" to "is the sensor behaving correctly / is the signal trustworthy":
一旦接上真实的 HU108，检查的重点就从“代码对不对”变成“传感器表现是否正常/信号是否可信”：

4.1 Noise floor at rest / 静止状态下的噪声水平

- Leave the sensor stationary on a table.
  把传感器静置在桌面上不动。
- Readings should hover close to 0, with only small random noise.
  读数应接近 0，只有很小的随机噪声。
- Bad sign: large swings (e.g. ±0.5) while completely still → wiring issue, grounding problem, or sensor fault.
  异常表现：完全静止时读数却大幅跳动（比如±0.5）→ 可能是接线问题、接地问题，或传感器本身故障。

4.2 Response to a known excitation / 对已知激励的响应

- Gently tap the surface the sensor is mounted on.
  轻敲传感器所在的表面。
- You should see a clear spike, followed by decay back to baseline.
  应该能看到明显的尖峰，然后衰减回基线水平。
- Bad sign: no reaction to a tap, or the signal keeps oscillating/diverging instead of settling → sampling or filtering issue.
  异常表现：敲击没有反应，或者信号持续震荡/发散而不收敛 → 可能是采样或滤波方面的问题。

4.3 Axis independence / 三轴是否独立

- Move/shake the sensor along one axis only (e.g. X).
  只沿一个方向（比如 X 轴）晃动传感器。
- Mainly the corresponding axis should show large changes; the others should stay relatively calm.
  主要应该是对应那个轴的读数变化大，其他轴应相对平稳。
- Bad sign: all three axes always move together → possible mounting orientation issue or a bug in the acquisition code.
  异常表现：三个轴总是同步一起变化 → 可能是安装方向问题，或采集代码有 bug。

4.4 Sampling stability / 采样稳定性

- Same check as in the simulated test — inspect timestamp intervals in the CSV for consistency.
  和模拟测试一样的检查方法——看 CSV 里的 timestamp 间隔是否一致。

4.5 Compare against the datasheet / 对照数据手册

- Measurement range, resolution, and noise density from the vendor datasheet should roughly bound what you observe.
  厂商数据手册里的量程、分辨率、噪声密度，应该能大致框住你实测出来的数值范围。
- This is why hardware/HU108/datasheet.md has TBD fields to fill in once the real spec is confirmed.
  这也是为什么 hardware/HU108/datasheet.md 里留了 TBD 字段，等确认了真实参数之后再填上。

5. Summary / 总结

Stage	Question answered
--simulate (Synthetic, designed by us / 人为设计的合成数据)	Does the pipeline (code) work correctly? 整条流程（代码）是否正常运行?
Real hardware (Actual HU108 output / 真实的 HU108 输出)	Is the sensor / signal trustworthy? 传感器/信号是否可信?

Don't judge sensor quality from simulated data, and don't judge code correctness only by "it produced some numbers" — check the specific signs above for each stage.
不要用模拟数据去判断传感器好坏，也不要仅凭“跑出了一些数字”就认为代码没问题——针对每个阶段，按上面列的具体检查项逐一确认。